REGULARITY OF A RING MAP TO A NOETHERIAN RING IS A LOCAL PROPERTY

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file more-algebra.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The chapter file pinned-original-source.tex carries only \input{preamble} on its first line, so no preamble text lives inside the chapter. The retained file sources/stacks-more-algebra/retained-preamble.tex is a mechanical, line-for-line copy of the project's single shared preamble as staged in the snapshot (Xy-pic setup plus theorem declarations and mathematical macros). Nothing was added, removed or redefined, and the excerpt serves only as preamble, never as a span coordinate.

Editorial note. Both counted spans start at the \begin line of their environment, so the statement and the author's proof are delivered whole; surrounding prose of the section is deliberately outside the counted spans and is carried as standard prerequisites instead.

Editorial note. The citation fields (source_id, source_version, authors, source_url, license.name, source_title) are identical to those of every other card this lane has produced from this chapter, so the batch selector treats them as one source.

Allowed standard prerequisites (editorial).

- A Noetherian ring is regular if and only if all its local rings are regular local rings. Algebra, Definition \ref{algebra-definition-regular}, cited at proof lines 10414-10416. Reduces regularity to the local rings.
- A ring map is flat if and only if all induced maps of local rings are flat. Algebra, Lemma \ref{algebra-lemma-flat-localization}, cited at proof lines 10416-10418. Reduces flatness to the local rings.

Source reference. Original lemma 07C0: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 9360–9360. The lemma's own canonical tag, quoted so that readers can resolve the label inside the delivered text; it points at the counted statement itself and carries no proof content.

Source reference. Original lemma 00OD: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 891–891. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Source reference. Original lemma 00HT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 655–655. Navigation pointer to a statement the counted proof refers to but does not contain; nothing load-bearing is delegated to it.

Lemma 0.2 (Regular is a local property)**.** *Let $R \to \Lambda$ be a ring map with Λ Noetherian. The following are equivalent*

(1) *$R \to \Lambda$ is regular,*

Date: stacks-project:more-algebra.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

(2) $R_{\mathfrak{p}} \to \Lambda_{\mathfrak{q}}$ *is regular for all* $\mathfrak{q} \subset \Lambda$ *lying over* $\mathfrak{p} \subset R$*, and*
(3) $R_{\mathfrak{m}} \to \Lambda_{\mathfrak{m}'}$ *is regular for all maximal ideals* $\mathfrak{m}' \subset \Lambda$ *lying over* $\mathfrak{m}$ *in* R*.*

Proof. This is true because a Noetherian ring is regular if and only if all the local rings are regular local rings, see Algebra, Definition 00OD and a ring map is flat if and only if all the induced maps of local rings are flat, see Algebra, Lemma 00HT. □